Cluster Sampling Design Learning Activity

STA304 — Summer 2026

— Solutions —

Background

Population data

Table 1: Weekly homework hours for all 30 students across 5 classrooms

Classroom (PSU)	S1	S2	S3	S4	S5	S6	Room mean
Room 1	4	6	5	7	3	5	5.00
Room 2	8	9	7	6	8	10	8.00
Room 3	2	3	4	2	5	3	3.17
Room 4	6	5	7	4	6	8	6.00
Room 5	10	9	11	8	10	12	10.00
Population mean	$\bar{y}_{\mathcal{U}} = 193/30 = 6.43$ hrs/week						

Given: $\widehat{\text{MSB}} = 14.2$, $\widehat{\text{MSW}} = 2.8$, $c_1 = 4$, $c_2 = 1$, $C = 20$, $N = 5$, $M = 6$.

$$V(\bar{y}) = \left(1 - \frac{n}{N}\right)\frac{\text{MSB}}{nM} + \left(1 - \frac{m}{M}\right)\frac{\text{MSW}}{nm}$$

Part A: Enumerate Feasible Designs

A. Derive $m_{\max}(n)$.

Solution. Starting from the cost constraint $c_1 n + c_2 nm \le C$, isolate m:

$$c_2 nm \le C - c_1 n \implies m \le \frac{C - c_1 n}{c_2 n} = \frac{20 - 4n}{n}$$

Since m must be a positive integer and cannot exceed the PSU size $M = 6$:

$$\boxed{m_{\max}(n) = \min\left(6,\ \left\lfloor \frac{20 - 4n}{n} \right\rfloor\right)}$$

B. Complete the feasibility table.

Solution. For each n, compute $m_{\max}$, cost, and variance terms using Equation (1).

Sample solutions for $n = 3$:

$$m_{\max} = \min\left(6,\ \left\lfloor \frac{20-12}{3} \right\rfloor\right) = \min(6,\ \lfloor 2.67 \rfloor) = \min(6, 2) = 2$$

$$\text{Cost} = 4(3) + 1(3)(2) = 12 + 6 = 18$$

$$\text{Term 1} = \left(1 - \frac{3}{5}\right)\frac{14.2}{3 \times 6} = \frac{2}{5} \cdot \frac{14.2}{18} = 0.4 \times 0.7889 = 0.3156$$

$$\text{Term 2} = \left(1 - \frac{2}{6}\right)\frac{2.8}{3 \times 2} = \frac{4}{6} \cdot \frac{2.8}{6} = 0.6\overline{6} \times 0.46\overline{6} = 0.3111$$

$$V(\bar{y}) = 0.3156 + 0.3111 = 0.6267$$

Table 2: Completed feasibility table

n	$m_{\max}$	Cost	Term 1	Term 2	$V(\bar{y})$	
1	6	10	1.8933	0.0000	1.8933	
2	6	20	0.7100	0.0000	0.7100	
3	2	18	0.3156	0.3111	**0.6267**	← minimum
4	1	20	0.1183	0.5833	0.7017	

Note on $n = 1$ and $n = 2$: When $m = M = 6$, the within-cluster PSU is fully sampled, so Term 2 $= (1 - 6/6)\,(\cdots) = 0$ exactly.

C. Which design minimizes $V(\bar{y})$?

Solution. The minimum variance is achieved at $n^* = 3$, $m^* = 2$, giving $V(\bar{y}) = 0.6267$.

Part B: Precision-Based Sample Size Determination

A. Required bound for $d = 2$ hr.

Solution.

$$V(\bar{y}) \leq \left(\frac{2.0}{1.96}\right)^2 = \mathbf{1.041233}$$

B. Does any feasible design achieve $d = 2$?

Solution. From the completed Table 2, the minimum achievable variance within the budget is $V = 0.6267$ at $(n^*, m^*) = (3, 2)$. Since $0.6267 < 1.041233$, this choice of n and m satisfies our budget and our desired precision.

C. Tightened requirement: $d = 1$ **hr.**

Solution.

$$V(\bar{y}) \le \left(\frac{1}{1.96}\right)^2 = (0.4082)^2 = \mathbf{0.260}$$

The minimum feasible variance is 0.6267 which exceeds 0.260. **This level of precision is also not achievable within the budget.**

Part C: Conduct the experiment

A. Generate a sample

Solution. Using a random number generator, Room 1, 4, and 5 were selected.

From room 1, S2 and S6 were selected. From room 4, S3 and S6 were selected. From room 5, S3 and S5 were selected

Observed sample:

Table 3: Observed sample of weekly homework hours

Classroom (PSU)	s1	s2	Room mean
Room 1	6	5	5.5
Room 4	7	8	7.5
Room 5	8	10	9

B. 95% confidence interval for the population mean

Solution. Upon reflection, I decided to use 1.96 for my critical value since that's what was used for the sample size calculation in Part B. But to be more precise, the sample size calculation should have been computed using the t-distribution. It would have been more complicated to do because both the variance and the t-distribution depends on n. You would have to use a computer to do this.

```
  data <- data.frame(
block_id = rep(1:3, each = 2),
hh_id    = rep(1:2, times = 3),
hours   = c(6,5,7,8,8,10),
N_psu = 5,
M_i      = 6)


design <- svydesign(
```

```
  id = ~block_id + hh_id,
  fpc = ~N_psu + M_i,
  data = data
)

svymean(~hours, design)

> svymean(~hours, design)
mean     SE
hours 7.3333 0.6912

> confint( svymean(~hours, design))
         2.5 %   97.5 %
hours 5.978577 8.688089
```

C. Discussion

Solution. The population mean is 6.43 and it is within the realized 95% confidence interval. The width of the confidence interval is 2.71 and the margin of error is half this, 1.355. This is between the desired precision.

Note here that the desired margin of error may not be achieved since you used a t-distribution (as instructed) and this will produce larger confidence interval with a small sample size. The sample size calculation in Part B was not done with a t-distribution so there is discrepancy. It would have been more complicated to do because both the variance and the t-distribution depends on n. You would have to use a computer to solve for n.

Also note, that I did not use the given estimated MSW and MSB from a previous study. In general, those are just meant as guidelines when you are designing a study. Once you conduct the study and collect the data, your analysis is generally done on the new observed data. Although there are methods to add previous studies results in your model as well...